Comparative Analysis of Algorithms on CEC 2017 Benchmarks

SDSC3004 Final Project

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Table of Contents

1. Introduction & Methodology
2. Algorithms
3. Results

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Table of Contents

1. Introduction & Methodology
2. Algorithms
3. Results

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CEC 2017 Selected Benchmarks

The Assigned Challenge:

- **F2 (Unimodal):** Tests sheer *exploitation speed* along a narrow, ill-conditioned ridge.
- **F10 (Multimodal):** Tests *global exploration* against thousands of deceptive local traps.
- **F17 (Hybrid):** Tests *structural robustness* by fracturing dimensions into conflicting sub-functions.
- **F27 (Composition):** The ultimate *reliability test*, blending multiple asymmetrical landscapes into chaotic terrain.

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Experimental Methodology

To ensure absolute statistical rigor, all algorithms were subjected to the exact same constraints:

- **Dimensionality:** $D = 30$ for all problems.
- **Statistical Robustness:** 10 independent runs per algorithm to account for stochastic variance. Results report the Mean and Min/Max margins.
- **The Metric of Fairness:** Function Evaluations (FEs). In the experiments, we set the maximum function evaluations as 300,000.

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Table of Contents

1 Introduction & Methodology

2 Algorithms

3 Results

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Selected Algorithms

- **Differential Evolution (DE)**
- **Self-Adaptive Differential Evolution (jDE)**
- **Simulated Annealing (SA)**
- **Dual Annealing (DA)**
- **Sephrical Search (SS)**
- **Self-Adaptive Sephrical Search (SASS)**

The Evolutionary Swarms

Differential Evolution (DE)

The Baseline Swarm

- **Mechanism:** Evolves a population using vector math: $V = x_1 + F \cdot (x_2 - x_3)$.
- **Search Style:** Uses the physical distance between population members to naturally scale search steps.
- **Fatal Flaw:** Requires the user to blindly guess the perfect Mutation Factor (F) and Crossover Rate (CR). Highly prone to premature convergence.

Self-Adaptive DE (jDE)

The Upgrade

- **Mechanism:** "Survival of the fittest parameters."
- **Search Style:** Attaches unique F and CR values to every single vector. Successful parameters are rewarded and passed to the next generation.
- **Advantage:** Automatically tunes itself to match the landscape (e.g., switches to aggressive exploration when trapped).

Thermodynamic Search

Simulated Annealing (SA)

The Classic Trajectory

- **Mechanism:** Inspired by cooling metal. A single agent moves across the map.
- **Search Style:** Early on (high temperature), it accepts "worse" moves to bounce out of fake valleys. As it cools, it becomes greedy. Accept "worse" with probability: $\exp(\frac{\Delta E}{T})$
- **Fatal Flaw:** Evaluating one point at a time is painfully slow.

Dual Annealing (DA)

The Global-Local Hybrid

- **Mechanism:** The "Global and Local" approach.
- **Global Phase :** Uses an advanced, generalized SA. Accept Probability: $P_a = \left[1 - (1 - q_a)\frac{\Delta E}{T}\right]^{\frac{1}{1-q_a}}$
- **Local Phase:** Deploys L-BFGS-B optimizer (Newton style). $X_{new} = X_{current} - \alpha B^{-1}\nabla f(X_{current})$

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Spherical Search (SS)

The Trap: Coordinate Bias

Standard algorithms search on a rigid X/Y grid. If a landscape is diagonal (like F2), they are forced to inefficiently "zigzag" down the canyon walls.

The Fix: Escaping the Grid

SS encases the search space inside a multi-dimensional sphere, bypassing the grid entirely.

The Geometric Engine:

- **Matrix Rotation:** It uses an orthogonal rotation matrix (Q) to mathematically tilt its search space to perfectly match the canyon's diagonal slope.
- **Fluid Movement:** New trial solutions are generated on the surface of this rotated sphere using a fixed mutation rate (c):

$$X_{trial} = X_{best} + c \cdot Q \cdot S$$

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Self-Adaptive Spherical Search (SASS)

The Limitation of SS

While pure Spherical Search has incredible geometric speed, its parameters (Mutation c and Crossover rp) are strictly fixed. On highly chaotic landscapes (like Composition F27), fixed parameters lead to trap vulnerability.

The SASS Upgrade: SASS upgrades the SS geometric engine with an adaptive ability:

- **Historical Memory:** SASS implements a SHADE-style memory matrix. It tracks which c and rp values successfully find better solutions and automatically shifts the swarm's parameters toward those winning values.
- **External Archive:** It stores failed parent vectors to maintain structural diversity, preventing premature convergence.

Result: SASS combines the geometric speed of a sphere with the survival instincts of a self-tuning parameter engine.

SASS: The Mathematical Engine

To achieve this 360-degree rotational freedom, SASS relies on three specific geometric steps:

1. **Spherical Boundary:** It defines a search center (X_{center}) and a dynamic radius (r) based on the current top-performing swarm members.
2. **Matrix Rotation:** It utilizes an **orthogonal rotation matrix** (Q) to tilt the search space, perfectly aligning its internal axis with the landscape's gradient.
3. **Vector Generation:** It generates new trial solutions on the surface of this rotated sphere using the core update formula:

$$X_{trial} = X_{center} + r \cdot Q \cdot S$$

*(*Where S is a randomly distributed vector on a unit hyper-sphere.)*

Impact: The Q matrix mathematically neutralizes rotation penalties (like those in F2 and F27), granting SASS unmatched exploitation efficiency.

Table of Contents

1 Introduction & Methodology

2 Algorithms

3 Results

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Comparison of Optimization Algorithms

Algorithm	Unimodal (NO2)	Multimodal (NO10)	Hybrid (NO17)	Composition (NO27)
DE	2165.17 / (613.3)	42082.7 / (15580)	2373.8 / (45.9)	3210.8 / (470.4)
jDE	**200.00 / (0.0)**	1245.6 / (26.7)	2054.3 / (37.3)	**2726.7 / (0.0)**
SA	6271.3 / (3549)	33480.7 / (10927)	100649.3 / (78818)	6009.8 / (2253)
DA	**200.00 / (0.0)**	1379.8 / (94.5)	**1735.6 / (27.9)**	**2746.2 / (0.0)**
SS	**200.00 / (0.0)**	51035.6 / (22682)	100995.3 / (58323)	**2700.0 / (0.0)**
SASS	**200.00 / (0.0)**	**1016.5 / (7.9)**	**1722.1 / (20.2)**	**2701.1 / (0.0)**

**Values shown as Mean Score / (Best Error Gap). Theoretical minimums subtracted.*

- **Observation:** SASS consistently achieves the lowest error across all benchmarks.
- **Efficiency:** jDE and DA are highly effective on simpler landscapes, but SASS dominates in Multimodal and Hybrid scenarios.

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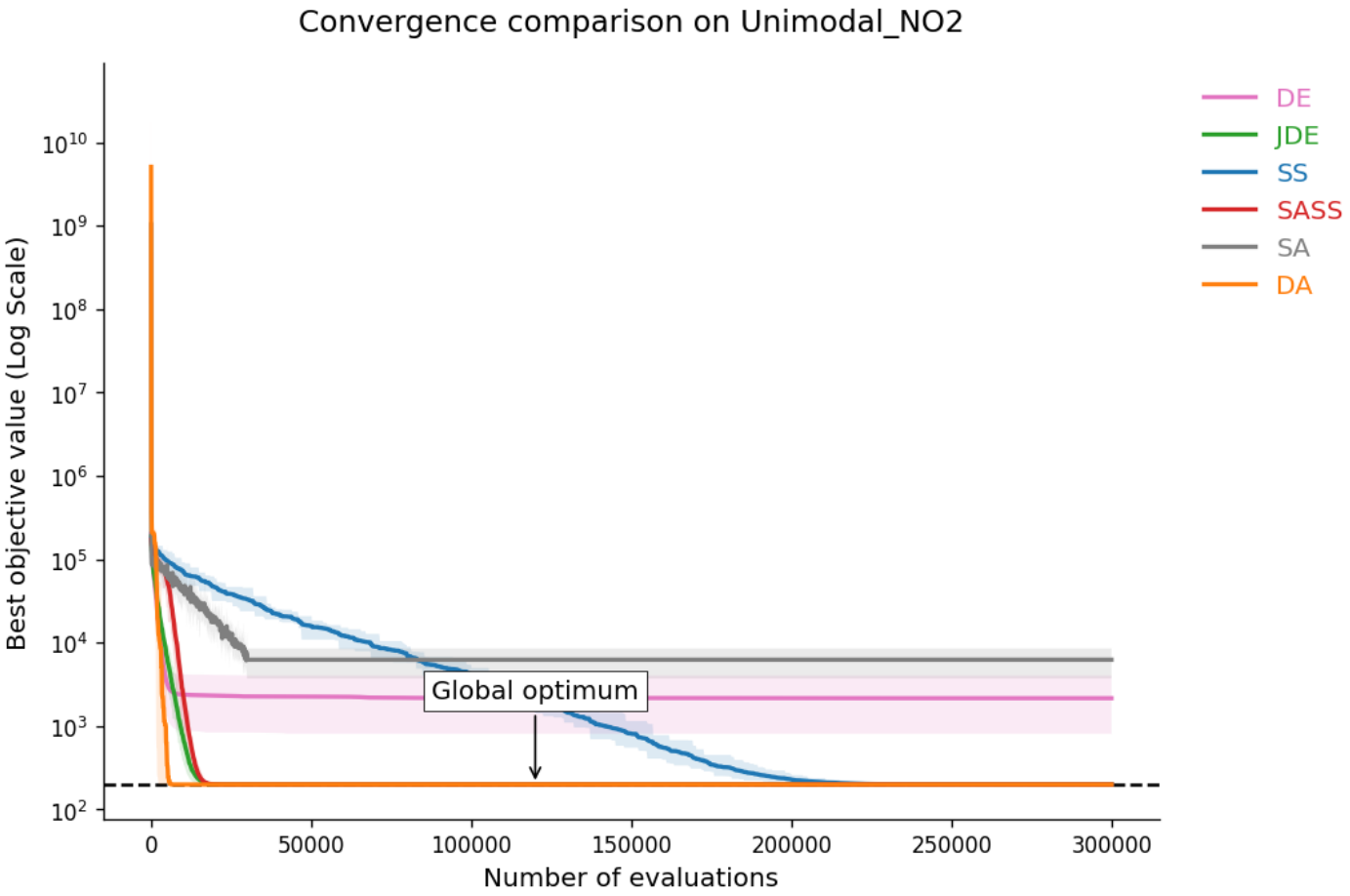
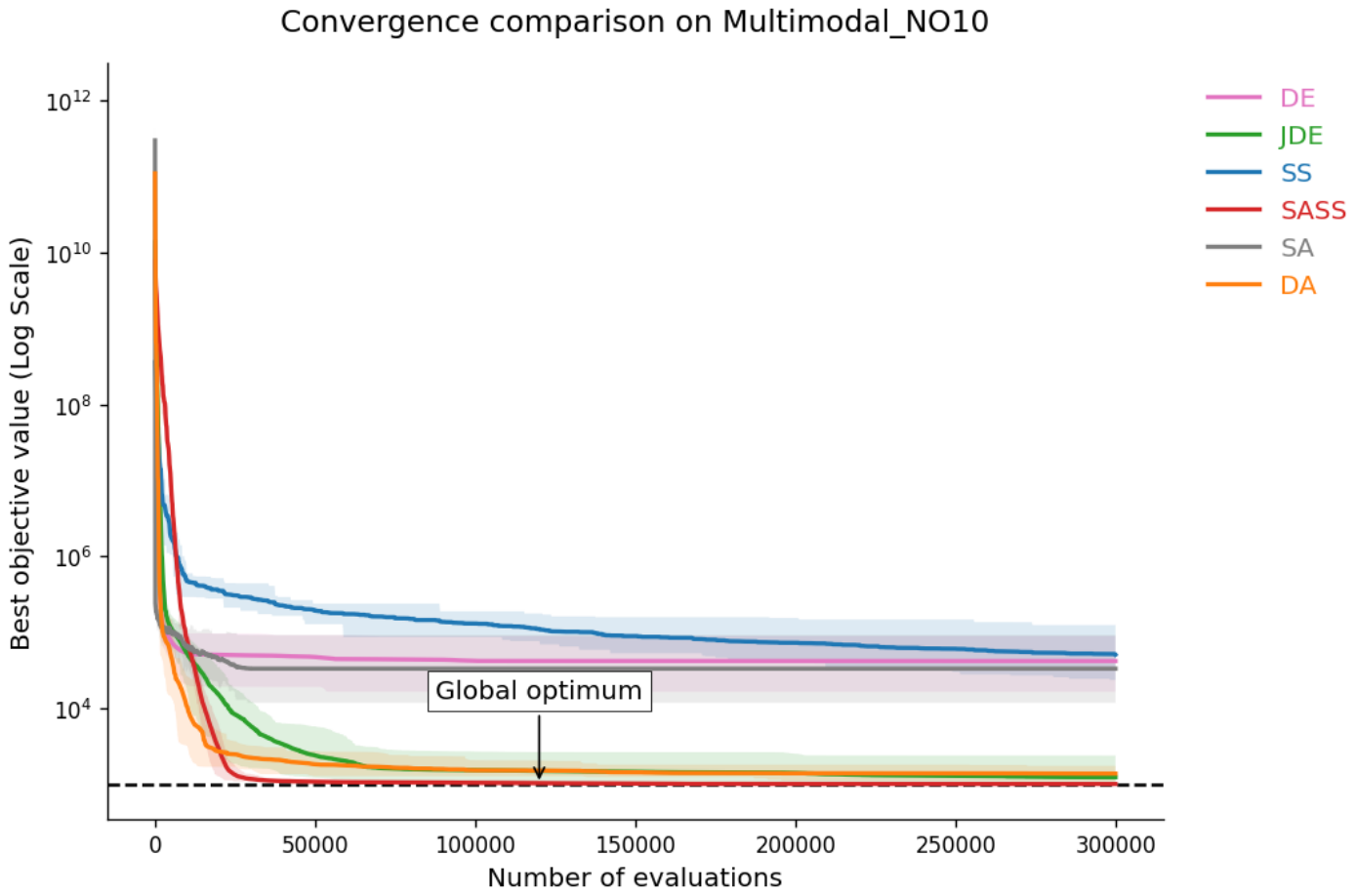
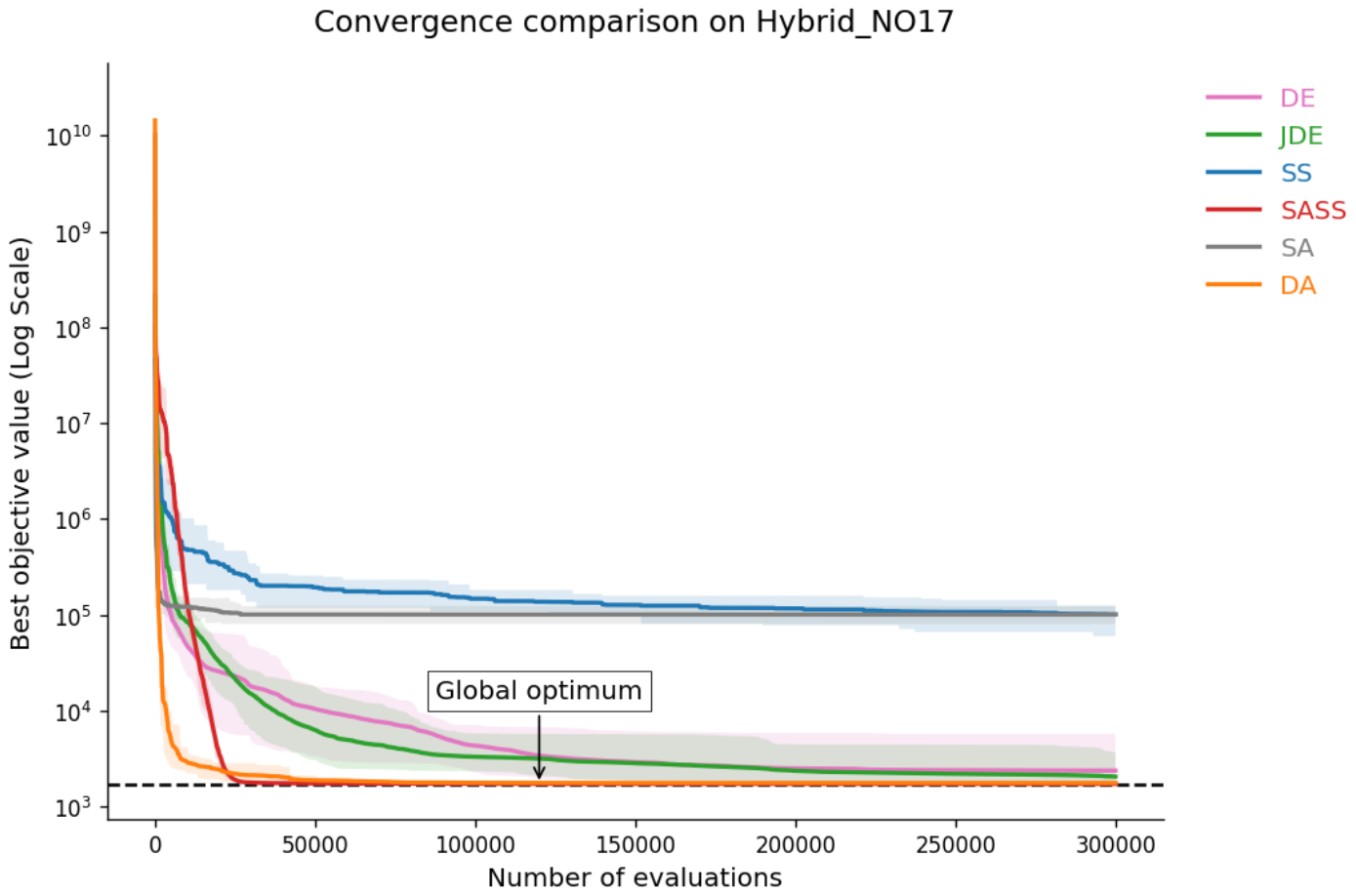
Result: Unimodal No.2



Result: Multimodal No.10



Result: Hybrid No.17



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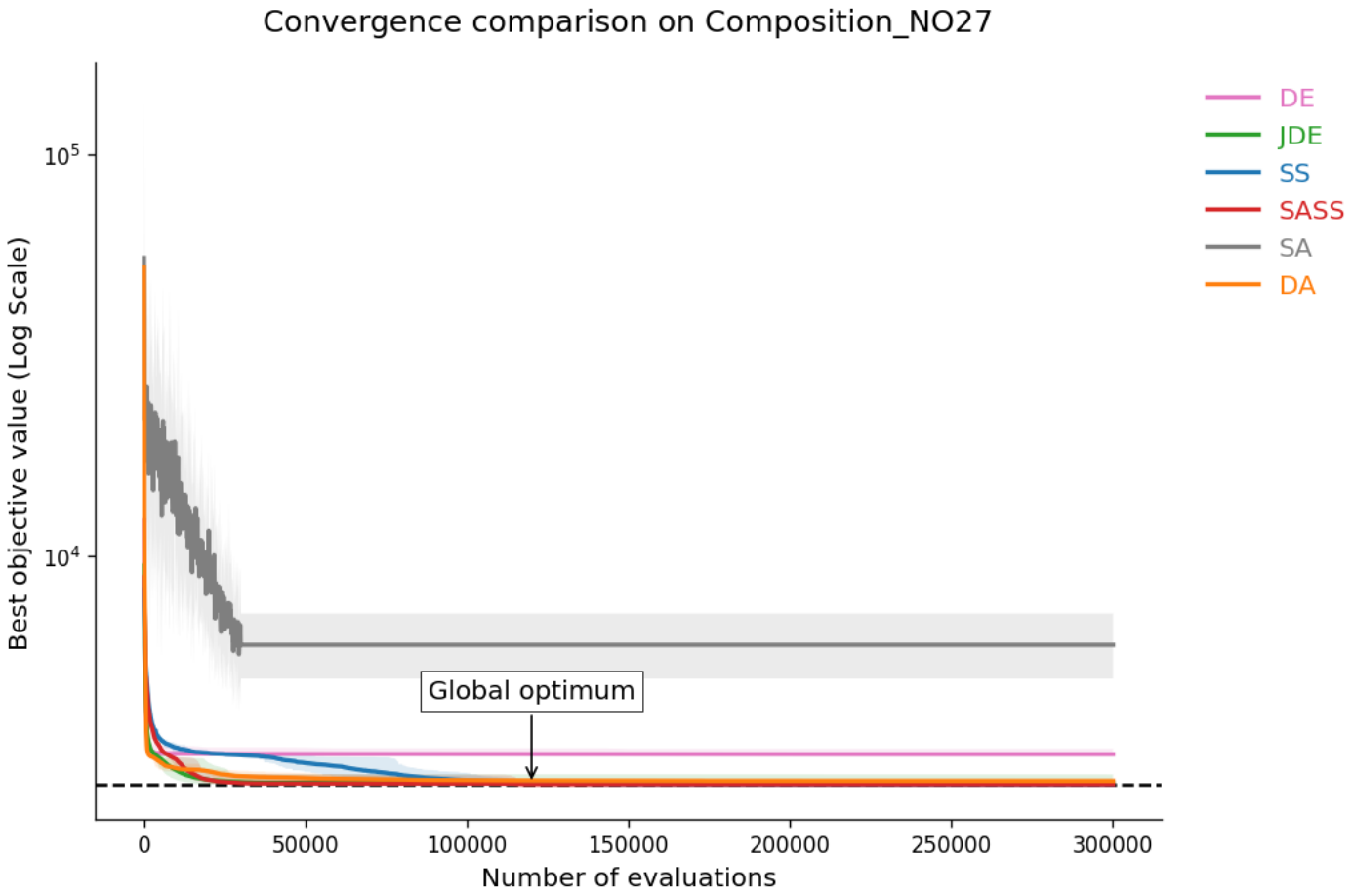
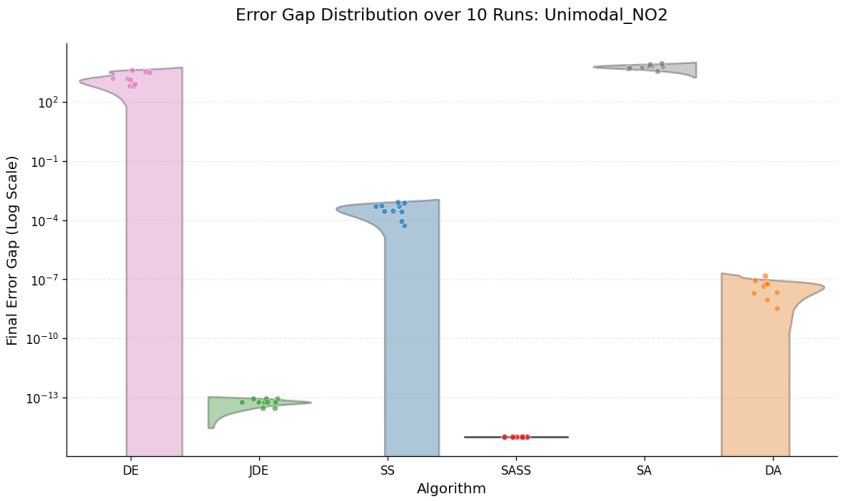
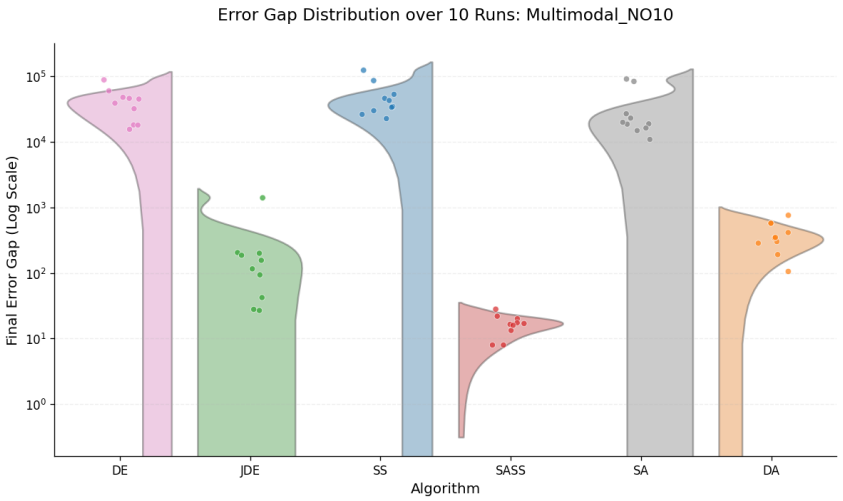
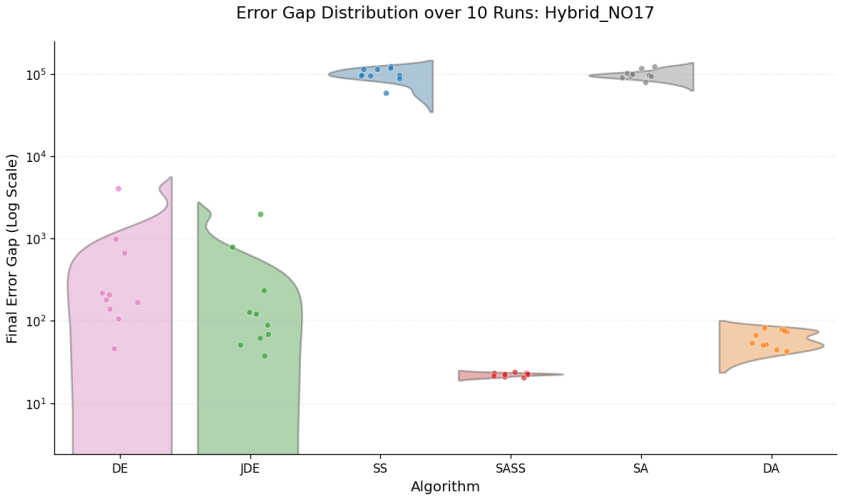
Result: Composition No.27

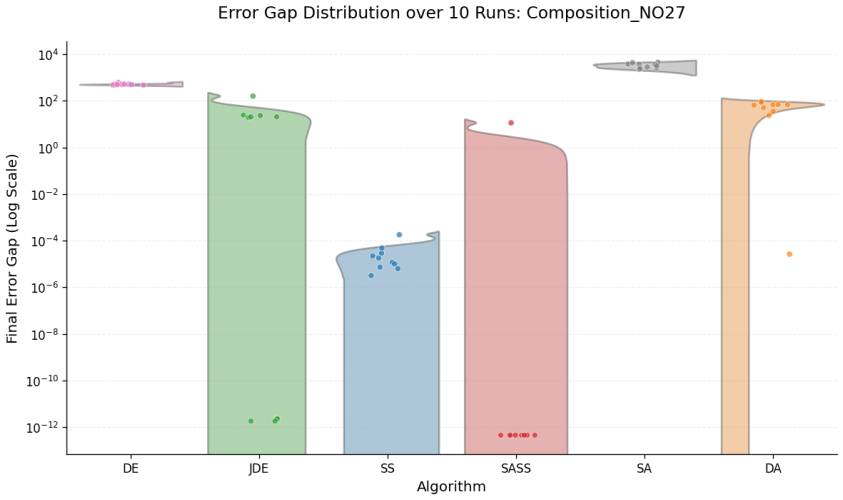


Result: Comparison on Stability









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Thank You!

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